VOLUME Theorems

Theorems of the Project

T1–T10: statements, lemmas, proofs and honest labels

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Hilbert Polya Bridge · monograph series C1–C8 · том Theorems · версия 1.0.0 · October 10, 2026

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About this volume

This volume collects the ten statements on which the whole series rests. Each carries a status label: proven — a complete proof plus a machine verification exists; measured — a statement about a finite computational experiment with controls; conjecture — a working hypothesis of the series. We deliberately do not inflate the status: the difference between a proof and a measurement is part of the method.

The numbering T1–T10 is shared by the READMEs, the monographs C1–C8, the Hilbert–Pólya scorecard and the Independent Verification Protocol. The proofs are given in a volume sufficient for checking without external sources; every constant is derived inside the series.

T1–T3: the frame and the counting

T1 (Γ-closures) — proven.

$|\Gamma(\tfrac{1}{2}+iu)|^2 = \pi/\cosh \pi u$; $|\Gamma(iu)|^2 = \pi/(u\cdot\sinh \pi u)$; $|\Gamma(\tfrac{1}{4}+iu)|\cdot|\Gamma(\tfrac{3}{4}+iu)| = \pi\sqrt{2}/\sqrt{\cosh 2\pi u}$; the single modulus on the line admits no elementary closure (K1, refuted).

Proof. *Euler reflection, Legendre duplication; the counterexample to the candidate form is 0.765 at $u = 1$. Verification at dps = 40: 1.3–2.6e-39.*

T2 (reality of Z_id) — proven.

Im Z_id(t) = 0 on the critical line; parity holds to 9.9e-29.

Proof. *$\xi(\tfrac{1}{2}+it) \in \mathbb{R}$ (the symmetry $\xi(s) = \xi(1-s)$), $\theta(t)$ odd. Verification at dps = 30.*

T3 (pole calibration) — proven.

$N(T) = \Delta\arg \Xi/2\pi$ with no additive +1 when the pole $s = 1$ is bypassed from above; a single leaf condition loses turns (2π aliasing).

Proof. *The residue of ξ'/ξ at $s = 1$ is $+1$; a clockwise half-passage contributes $-\pi$ to $\Delta\arg$. Practical verification: anchors 0/3/10/29/79/649 and the arithmetic bridge 2.6e-26.*

T4–T6: arithmetic and the trace

T4 (co-location) — proven.

$(\Phi_p, \Phi_q) \equiv \Phi_{pq}$ with $\Phi_p = \tfrac{1}{2}\ln p$; $\max|\Delta E| = 1.35e\text{-}14$.

Proof. *Additivity of the logarithm; the lattice realisation preserves the identity at machine level.*

T5 (resolution kernel) — measured.

The composite-node response $\delta(d)$: 0 → 3.2e-3 (d = 0.5) → 1.5e-2 (d = 2) → saturation ~2e-2 at $d \approx 4$–6; multiplicativity is localised in 2–3 cells; first-order additivity fails (~1e-2).

Proof. *Direct measurement on the grid $d \in [0, 6]$; honesty label: depends on the flow conventions of C6.*

T6 (trace formula for H_ξ) — proven and verified.

$Q(a) = \zeta'/\zeta(\frac{1}{2}+a) + 1/(\frac{1}{2}+a) + 1/(a-\frac{1}{2}) - \frac{1}{2}\ln\pi + \frac{1}{2}\psi(\frac{1}{4}+a/2) - B - C_0$ with $B = \frac{1}{2}\ln(4\pi) - 1 - \gamma/2 = -C_0$; $K(x) = \mathrm{Tr}\, e^{-xH_\xi} = (1/\pi)\int_0^\infty Q(a)\sin(ax)da$; for $a > \frac{1}{2}$ the ζ-series contributes the weight $\Lambda(n)/\sqrt{n}$.

Proof. *Hadamard product plus $2a/(a^2+\gamma^2) = 2\int e^{-\gamma t}\sin(at)dt$. Verification: $|\Delta Q| \le 1.7e\text{-}12$ (M0), $|\Delta K| \le 1.2e\text{-}5$ (M1).*

T7–T10: spectrum and symmetry

T7 (universality demarcation) — measured.

87 gates (52 protocol + 35 extended) lie in the GUE class ($\langle r\rangle = 0.498$–0.612) and carry no Λ-weights: $r_{H7} \le 0.226$, $|r_{H7}| \le 0.043$ over the new handles, r_{EF} inside the null clouds.

Proof. *Campaigns C3–C5 with positive controls PC1/PC2 ($r = 1.0$) and PC0 ($r = 0.9998$).*

T8 (64→4 reduction) — proven.

The E2 spin structures collapse to 4 Hamiltonians by the pairs (φ_x, φ_y) mod 2π; isospectrality within classes 1.1e-14.

Proof. *The phases enter through two sums of three 1-forms mod $2\pi \to 2\times 2 = 4$ classes; verified against the E1/E2 tables.*

T9 (zero modes at α = 1/3) — measured.

On frustrated tori $L \equiv 0 \pmod 4$ the clean lattice at $\alpha = 1/3$ has exact zero modes: 18 at $L = 12$, 42 at $L = 24$.

Proof. *Direct diagonalisation with fixed multiplicities; a number-theoretic explanation (the Dirac index) is an open question of the series.*

T10 (statistics ladder) — measured.

Poisson/GOE/GUE are governed by symmetry (frustration / real disorder / complex vortices); GSE is structurally unreachable ($T^2 = +1$).

Proof. *An 11-channel ladder, classification by $\langle r\rangle$ + KS; the GSE limit is architectural.*

Summary table and consequences

Of the ten statements, four are fully proven (T1, T2, T4, T6), one is proven and verified to a numerical floor (T6), four are measured with controls (T5, T7, T9, T10), and one is proven at the level of phase structure (T8). None carries a status higher than its content — and it is exactly this discipline that lets the series speak of the H2 null-level verdict (gates indistinguishable from surrogates) as a measured fact rather than an estimate.

The main consequences for the Hilbert–Pólya programme are collected in the scorecard: the resonator is built and tuned (H1+H3+H4), the zeros are encoded rather than generated (H2+H5), and the prime weight is verified only for the zero operator (T6). A hypothetical HP-operator, if it exists, is not a local deformation of

the lattice family — the single bridge between all ten theorems and the three projections of the gap.

Table 1. Statuses of the theorems

T	Content (brief)	Status
T1	Γ-closures: 3 forms + a refutation	proven
T2	Im Z_id = 0	proven
T3	pole calibration of the argument principle	proven
T4	co-location (Φ_p,Φ_q) ≡ Φ_pq	proven
T5	resolution kernel δ(d), ξ_mult ≈ 2–3	measured
T6	trace formula Tr e^{−xH_ξ}	proven+verified
T7	universality demarcation (87 gates)	measured
T8	64→4 reduction	proven
T9	zero modes at α = 1/3	measured
T10	Poisson/GOE/GUE ladder	measured